CPSY 1291 · Recitation handout 7

# Exam 1 review

Lectures 1–8 and the four steps of backpropagation, in sixteen problems

Fall 2026 · Week 7 · Exam 1 is in class, Thu 10/22

## How to use this

This is a review, not new material. Everything here has appeared in a lecture, an assignment, or an earlier recitation. Exam 1 covers Lectures 1–8 and the four steps of backpropagation from Lecture 9; nothing else from Lecture 9 and nothing after it.

Work in this order, and do not reverse it. **First** read Section 1, which is the whole of that material compressed into the statements you should be able to make without notes. **Then** do the problems in Section 2 with the summary closed. Reading a summary produces the feeling of understanding; answering a question produces the fact of it, and the gap between those two is where exam marks are lost.

> **What the exam can ask.** Every lecture in this course does three things: introduces an *object* (a distance, a matrix, a score, a rule), states *what it is for*, and states *when it misleads you*. Exam questions come from all three, and the third is the one students prepare least and are asked about most.

## 1 Lectures 1–8, compressed

### 1.1 L01 — Representations, spaces and metrics

A representation is a pattern that carries information about a stimulus. Measured, it is a vector $\mathbf{x} \in \mathbb{R}^D$: 784 pixel values, 210 voxels, 482 IT firing rates, 4,096 network activations. $D$ is the number of measurements, not the number of objects, and each image is one point in that $D$-dimensional space. Stack $N$ images as rows and you have the **response matrix** $\mathbf{X} \in \mathbb{R}^{N \times D}$, whose entry $x_{ij}$ is feature $j$ of image $i$.

Single trials are noisy: two presentations of one image correlate at about 0.17 in monkey IT, and the median is 0.14 for fMRI. Averaging repeats keeps what repeats, so every response vector in this course is a trial mean. The **split-half correlation** — the mean of half the repeats against the mean of the other half (0.69 in the IT data) — says how reliable that mean is.

Pixels, voxels, neurons and units cannot be compared entry by entry: the vectors differ in length and in units. Their pairwise dissimilarities can. The **RDM D** $\in \mathbb{R}^{N \times N}$ holds $d_{ik} = d(\mathbf{x}^{(i)}, \mathbf{x}^{(k)})$, with a zero diagonal and $d_{ik} = d_{ki}$. Choosing $d$ is a scientific decision.

### 1.2 L02 — Similarity, MDS and comparing RDMs

| Measure | Computes | Ignores |
|---|---|---|
| Euclidean | length of the difference vector | nothing: keeps gain and offset |
| cosine | $1 - \cos\phi$ | gain (overall scale) |
| correlation | $1 - r$: cosine on centered vectors | gain *and* offset |

Choose by what should not count. With two components, $r = \pm 1$ always, so correlation needs more than two. A distance in the full sense (a metric) satisfies nonnegativity, identity, symmetry and the triangle inequality. Euclidean distance does; cosine and correlation fail identity (a brighter copy of an image sits at cosine distance 0 from it) and the triangle inequality.

**MDS**: given $\mathbf{D}$, find coordinates $\mathbf{Y} \in \mathbb{R}^{N \times m}$ whose distances $\hat{d}_{ik}$ reproduce $d_{ik}$, by minimizing the stress $\sum_{i<k}(\hat{d}_{ik} - d_{ik})^2$. Report **Stress-1**, which does not depend on the units (Kruskal: 0.05 excellent, 0.10 fair, 0.20 poor), and compare it with the stress of a shuffled matrix. Three warnings: the solution is not unique (shift, rotate or reflect it and nothing changes, so the axes mean nothing); $m$ is your choice (look for the elbow of stress against $m$); low stress is not truth (enough dimensions fit noise). **Non-metric MDS** fits only the rank order of the dissimilarities, which suits ratings: on the Assignment 1 ratings Stress-1 falls from 0.31 to 0.23.

**Shepard's law**: in the psychological space recovered by non-metric MDS, generalization falls exponentially with distance, $s = e^{-c\,d}$. The law is not circular, because only the order was fitted. The fair rival is the Gaussian $a\,e^{-bd^2}$, with the same number of parameters but flat at $d = 0$, so the test is at the origin. Against a network's distances the answer changes: that is a different ruler.

**Comparing two RDMs** (RSA): flatten each upper triangle, `np.triu_indices(N, k=1)`, $N(N-1)/2$ numbers, and take Spearman's $\rho$, which asks only whether the two systems order the pairs alike. $\rho$ alone means little: it needs a ceiling and a second model. Matching geometry is evidence about a representation, not a demonstration of shared mechanism.

### 1.3 L03 — PCA and eigendecomposition

Center each feature. The covariance matrix $\mathbf{C}$ has the variances on its diagonal and summarizes the spread and orientation of the points. Project a centered point onto a unit direction $\mathbf{w}$: $z = \mathbf{w}^\top \tilde{\mathbf{x}}$, and across stimuli $\mathrm{Var}(z) = \mathbf{w}^\top \mathbf{C} \mathbf{w}$. Projected variance and squared reconstruction error add to a fixed total, so maximizing one minimizes the other.

Most directions are turned by $\mathbf{C}$. An **eigenvector** is not: $\mathbf{C}\mathbf{v}_j = \lambda_j \mathbf{v}_j$, and its eigenvalue $\lambda_j$ is the variance along it. PCA is the eigendecomposition $\mathbf{C} = \mathbf{V}\mathbf{\Lambda}\mathbf{V}^\top$; PC1 is the eigenvector with the largest eigenvalue. The **scores** $z_j = \mathbf{v}_j^\top(\mathbf{x} - \boldsymbol{\mu})$ are the new coordinates, and $\hat{\mathbf{x}} = \boldsymbol{\mu} + \sum_{j \le k} z_j \mathbf{v}_j$ is the reconstruction. The scree plot and the cumulative curve give the **embedding dimension** at a chosen threshold; a variance threshold is a reconstruction budget, not a statement about any task. `PCA.fit_transform` centers each feature but does not z-score it.

Three cautions. **Scale**: with one feature in millimetres and one in metres, PC1 is $1.00\,x_1 +$

$0.00\,x_2$; after z-scoring it is $0.71\,x_1 + 0.71\,x_2$. Standardizing is a modelling decision, not a default. **Most variance is not most useful**: the largest direction in the 120 photographs (29%) is mostly brightness, and in the toy example PC2 carries 2.2% of the variance yet is the only direction that separates the groups. **PCA is flat**: a curved path needs more linear coordinates than it has degrees of freedom. In Bao et al., PCA of a network's responses gave an object space whose directions are hypotheses, tested on held-out images.

### 1.4 L04 — Nonlinear dimensionality reduction

A **manifold** is a curved surface that is flat up close. Three counts of dimension answer three questions: **ambient** (coordinates per point), **embedding** (the flat subspace PCA needs at a variance threshold) and **intrinsic** (dimensions needed to locate a point on the manifold). For the Swiss roll they are 3, 3 and 2. PCA overestimates the intrinsic dimension whenever the manifold curves, and PCA and MDS both fold the roll, because points on neighboring turns are close through the air.

**Isomap** measures along the sheet — shortest paths through a nearest-neighbor graph — then runs MDS. One edge between turns, or a sparsely sampled sheet, breaks it.

**$t$-SNE**: (1) in the data, a Gaussian around each point gives neighbor probabilities $P$, its width set by the **perplexity**, roughly how many neighbors each point listens to (typically 5–50, always below $N$); (2) in the map, a heavy-tailed Student-$t$ gives $Q$, which leaves room for crowded neighbors; (3) gradient descent on the KL divergence moves the map until $Q$ matches $P$. Close pairs dominate the mismatch, so $t$-SNE keeps neighbors, not distances: cluster sizes and gaps are not measurements, low perplexity makes clumps from noise, and another seed gives another map (PCA initialization is more repeatable). **UMAP** builds a neighbor graph too (`n_neighbors`, `min_dist`) and carries the same warnings. Sweep the settings and the seed, keep what holds, and remember that stability is not fidelity: a map can repeat itself and still disagree with the original space. A map suggests a question; the response space answers it.

### 1.5 L05 — Geometry of neural representations

**Rank** is the number of directions with nonzero variance. Noise makes every real recording full rank (480 IT neurons: rank 480), so look at how the variance spreads instead. Mouse V1 has no elbow: 90% of the variance needs 744 components, and the spectrum is a power law with slope near 1. The **participation ratio**

$$\mathrm{PR} = \frac{\left(\sum_j \lambda_j\right)^2}{\sum_j \lambda_j^2}, \qquad 1 \le \mathrm{PR} \le D,$$

needs no threshold: if $k$ directions share the variance equally, $\mathrm{PR} = k$. It counts how evenly the variance is spread, not how much information there is. z-scoring each unit changes it (32 raw against 401 z-scored for one simulated population), so report which you used.

A **linear readout** answers "yes" when $\mathbf{w}^\top \mathbf{x} + b > 0$; its boundary is a hyperplane, perpendicular to $\mathbf{w}$. In IT, animacy is read out for 93% of held-out views, against 50% for guessing

and 63% for the **majority-class baseline**. Two classes are **linearly separable** when one hyperplane puts them on opposite sides. XOR is not (75% at best); a third unit $x_1x_2$ lifts the points so that one plane separates them.

### 1.6 L06 — Reading and comparing representations

Two properties pull against each other. A representation is **flexible** if almost any grouping of the stimuli can be read out — measured by **shattering** — and **abstract** if a readout learned on some stimuli still works on stimuli it never saw — measured by **CCGP**.

$N$ stimuli in general position in $N-1$ dimensions can be split every way: three points in two units give 8 of 8 splits, three on a line 6 of 8, four at the corners of a square 14 of 16 (the XOR splits fail). Capacity is measured with **random labels**: a linear readout memorizes about two per dimension, and deep networks memorize ImageNet with shuffled labels (Zhang et al. 2017) while staying at chance on new photographs.

A unit whose responses are a weighted sum of the others never raises the rank. A nonlinear unit — the product $x_1x_2$, or an AND — does, and then all 16 splits are separable. That is why **nonlinear mixed selectivity**, common in prefrontal cortex, makes a population high-dimensional: its shattering dimensionality is close to the maximum of 24, against at most 8 for pure selectivity, and it drops on error trials.

**CCGP**: fit a readout on some conditions (dogs, chairs) and test it, weights fixed, on others (cats, tables). It is high when the new stimuli differ from the old only along directions the readout ignores. Of Bernardi et al.'s four toy geometries, points at random are flexible but not abstract, clusters abstract but not flexible, a square abstract but not fully flexible, and a bent square both. Hippocampus looks like the bent square: context CCGP 0.96 with high shattering.

Comparing two systems: **RSA** builds an RDM per system and compares rank orders (Spearman); **CKA** builds a Gram matrix (dot products of centered units) and compares values, and is unchanged by uniform rescaling or rotation of the units. CKA is more common for network layers, RSA for brains and judgments.

### 1.7 L07 — Geometry, baselines and tests

On error trials, context CCGP drops but decoding of context (all conditions in training) does not: the geometry breaks, the information does not.

A score means something only after three questions. **Is it above chance?** Unrelated data correlate a little by luck. A **permutation test** shuffles one system's image order (or a readout's labels) 1,000 times and asks whether the score beats the 95th percentile of the shuffled scores (IT against ResNet-18: 0.71 against 0.055). With more neurons than training images, any labels fit (480 neurons, 102 images: 1.00 on training, chance on test), and with few stimuli even unrelated systems reach a linear CKA near 1. **Would it hold on new images?** Cross-validation tests every image once with a fit that never saw it; **leakage** — letting test images into any step, from the z-scoring mean to the choice of units — makes the score optimistic, so split first and fit every step on the training images. A bootstrap over images says whether a ranking of models would survive another sample of images. **How high could**

**it be?** Noise caps every score; the **noise ceiling** comes from split-half reliability, and a score is read against it (0.3 is near the top if the ceiling is 0.35), alongside the shuffled floor and the same area in another brain. Finally, change the measure and the winner can change: RSA puts CLIP first, CKA ResNet-18.

### 1.8 L08 — Learning the weights

The model neuron keeps what a point neuron does and drops the rest: inputs $x_i$ (firing rates), weights $w_i$ (synaptic strengths, excitatory or inhibitory), the pre-activation $z = \mathbf{w}^\top \mathbf{x} + b$, and the output $a = g(z)$, most often a ReLU. Because $\mathbf{w}^\top \mathbf{x} = \|\mathbf{w}\|\|\mathbf{x}\| \cos\phi$, the neuron asks how much of **w** is in the input. Read as a **boundary**, **w** is perpendicular to the line $z = 0$ and $b$ slides it: one neuron is a linear classifier, the same readout as in L05–L07. Read as a **template**, **w** drawn as an image looks like what the neuron detects.

Learning needs a **loss** and **gradient descent**, $\mathbf{w} \leftarrow \mathbf{w} - \eta \nabla \ell$: the gradient points uphill, and the learning rate $\eta$ sets the step.

| Rule | Output $\hat{y}$ | Loss | Update |
|---|---|---|---|
| Perceptron (1958) | threshold | none: a heuristic | moves **w** only on mistakes |
| Delta rule (1960) | $\mathbf{w}^\top \mathbf{x}$ | $\frac{1}{2}(y - \hat{y})^2$ | $\Delta \mathbf{w} = \eta\,(y - \hat{y})\,\mathbf{x}$ |
| Logistic regression | $\sigma(z)$ | cross-entropy | $\Delta \mathbf{w} = \eta\,(y - \hat{p})\,\mathbf{x}$ |

The perceptron is the delta rule with a thresholded output, so its error is zero unless it is wrong; if a line separates the classes, it stops. The cross-entropy is 0 when the model is certain and right, 0.69 for a 50/50 guess, and 2.30 when it gives the correct class 0.1. One recipe throughout: **learning rate** $\times$ **error** $\times$ **input**.

### 1.9 From L09 — the four steps of backpropagation

A hidden unit has no target, so its error has to come from the output's. With a sigmoid output and the cross-entropy:

1. **Forward**: run the input through and keep every weighted sum $z$ and every output $a$.
2. **Output error**: $\delta_{\text{out}} = \partial \ell / \partial z_{\text{out}} = \hat{y} - y$.
3. **Backward**: each hidden unit's error is the errors of the units it feeds, each times the weight to them, all times its own slope, $\delta_j = g'(z_j) \sum_k w_{kj}\, \delta_k$. A ReLU's slope is 1 if it was on and 0 if it was off, so a unit that was off gets no error.
4. **Gradients**: each weight's gradient is the error of the unit it enters times its input, $\partial \ell / \partial w = \delta \times$ input; each bias gets its unit's error.

The lecture's 2–2–1 XOR network, input $(1, 0)$, target 1: $z^{[1]} = (0.50, 0.30)$, $z^{[2]} = 0.17$, $\hat{y} = 0.54$, loss 0.61; $\delta^{[2]} = -0.46$; $\delta^{[1]} = (-0.32,\ 0.27)$; $\partial \ell / \partial w_1^{[2]} = -0.46 \times 0.50 = -0.23$. One step with $\eta = 1$ raises $\hat{y}$ to 0.82.

## 2 Problems

Do these closed-book. Answers follow.

**1.** A unit's activity is $(2, 4, 6)$ across three stimuli; another's is $(1, 2, 3)$. Give the cosine distance and the correlation distance between them, and explain the pair of answers in one sentence.

**2.** Why is correlation distance undefined for a unit that produces the same value for every stimulus?

**3.** You are told $d(A, B) = 0.2$, $d(B, C) = 0.2$, $d(A, C) = 0.9$. Which requirement of a distance fails, and name a measure from this course that permits it.

**4.** Someone reports that an MDS map's horizontal axis is "animacy". What is the methodological objection, and what evidence would answer it?

**5.** Two features are recorded, one in millimetres and one in metres. What is PC1 likely to be in raw units, what changes after z-scoring, and when should you not z-score?

**6.** PC1 keeps 97.8% of the variance and PC2 2.2%. Should you drop PC2 before training a readout to separate two groups?

**7.** Give the ambient, embedding and intrinsic dimension of a Swiss roll (a 2-D sheet rolled up in 3-D), and say which of them PCA measures.

**8.** Two $t$-SNE runs on identical data, differing only in `random_state`, give visibly different maps. Is either wrong? What should be reported, and how would you check a grouping?

**9.** Four units have covariance eigenvalues $(4, 4, 0, 0)$; another four have $(7, 1, 0, 0)$. Give the rank and the participation ratio of each, and say what the difference means.

**10.** Four stimuli sit at the corners of a square in two units. How many of the 16 splits can a linear readout separate, which fail, and what kind of third unit fixes them?

**11.** On error trials, context CCGP drops but decoding of context does not. What has changed in the representation, and what has not?

**12.** A readout on 480 IT neurons, trained on 102 images with shuffled labels, scores 1.00 on its training images. Explain, and say what you report instead.

**13.** You z-score your responses using the mean and standard deviation of all images, then run cross-validation. What is wrong, and how large is the problem compared with choosing units by their test score?

**14.** Delta rule: $\mathbf{w} = (0.5, -1)$, $b = 0$, $\mathbf{x} = (2, 1)$, $y = 1$, $\eta = 0.1$. Compute $\hat{y}$, $\Delta\mathbf{w}$, and $\hat{y}$ after the update.

**15.** A logistic-regression neuron gives $\hat{p} = 0.5$ for an example of class 1, and another gives $\hat{p} = 0.1$. Give each cross-entropy, and the update $\Delta\mathbf{w}$ in terms of $\mathbf{x}$.

**16.** The lecture's XOR network with its starting weights (hidden $\mathbf{w}_1 = (0.50, -0.40)$, $\mathbf{w}_2 = (0.30, 0.80)$, output weights $(0.70, -0.60)$, biases 0, ReLU hidden units, sigmoid output, cross-entropy). Input $(1, 1)$, target 0 (XOR of 1 and 1). Run the four steps of backpropagation and give every gradient.

### Answers

**1.** Cosine distance $0$: the two vectors point in exactly the same direction, differing only by a factor of two. Correlation distance $0$ as well: after centering, both become proportional to $(-1, 0, 1)$. The units differ only in gain, which cosine discards and correlation discards along with offset, so neither can tell them apart; Euclidean distance keeps gain and can.

**2.** Centering a constant vector gives the zero vector, whose length is zero, and the correlation divides by that length. In code it surfaces as `nan`, which then propagates silently through the whole matrix.

**3.** The triangle inequality: $0.9 > 0.2 + 0.2$. Cosine distance and correlation distance both permit it; Euclidean distance does not.

**4.** The MDS solution can be shifted, rotated or reflected without changing any distance or the stress, so "the horizontal axis" is not a property of the data; only relative positions are. Answering it requires independent evidence: relate a direction in the map to an external animacy rating, or show that a readout for animacy works on held-out stimuli.

**5.** PCA maximizes variance, so the feature in millimetres dominates: PC1 is close to $1.00\,x_1 + 0.00\,x_2$ and the second feature is invisible. After z-scoring, PC1 weights both (0.71 each in the lecture's example) and recovers the structure they share. Do not z-score by default when the units are comparable and the variances differ for real reasons: standardizing changes the relative weighting of the features.

**6.** No. PCA ranks directions by variance; the task decides what is useful. In the lecture's example both groups have the same PC1 coordinates, and a threshold at zero on PC2 separates them.

**7.** Ambient 3, embedding 3, intrinsic 2. PCA measures the embedding dimension, a linear count that needs a variance threshold, and overestimates the intrinsic dimension whenever the manifold curves.

**8.** Neither is wrong: the random start is set by the seed. Report what is stable across seeds and across a perplexity sweep, and say how many runs you made. Sizes and gaps in the map are not distances in the data; to check a grouping, compare each image's nearest neighbors in the original response space, or test a readout on held-out images.

**9.** Both have rank 2. PR $= 8^2/(16+16) = 2$ for the first and $8^2/(49+1) = 1.28$ for the second. Both use two directions, but in the second one direction carries almost all the variance; PR counts how evenly the variance is spread, not how much information the population carries.

**10.** 14 of 16. The two XOR splits, which put opposite corners together, fail. A third unit whose responses are not a weighted sum of the first two — $x_1 x_2$, or an AND unit — raises the rank to 3, and all 16 become separable. Adding $x_1 + x_2$ leaves the rank at 2.

**11.** The abstract format of context has broken: a readout trained on some conditions no longer transfers to the others. The information has not gone: a readout trained on all conditions still decodes context.

**12.** With more neurons than training images, the points are nearly perpendicular and any

labelling can be fit; scoring 1.00 means memorizing. Report accuracy on held-out images, against the shuffled-label score on test images (0.56 in the lecture, chance) and the majority-class baseline.

**13.** Leakage: the test images influenced the z-scoring, so the test score is somewhat optimistic. It is a small leak. Choosing units or settings by their test score is a large one — even pure noise can then look predictive. Split first, then fit every step on the training images.

**14.** $\hat{y} = 0.5 \cdot 2 - 1 \cdot 1 = 0$. Error $y - \hat{y} = 1$, so $\Delta\mathbf{w} = 0.1 \times 1 \times (2, 1) = (0.2, 0.1)$ and $\mathbf{w}$ becomes $(0.7, -0.9)$. After the update $\hat{y} = 1.4 - 0.9 = 0.5$, and the squared-error loss falls from 0.5 to 0.125.

**15.** $-\log 0.5 = 0.69$ and $-\log 0.1 = 2.30$. The update is $\Delta\mathbf{w} = \eta\,(y - \hat{p})\,\mathbf{x}$: $0.5\,\eta\,\mathbf{x}$ for the first and $0.9\,\eta\,\mathbf{x}$ for the second, so the more confidently wrong neuron moves more.

**16.** *Step 1.* $z_1^{[1]} = 0.50 - 0.40 = 0.10$ and $z_2^{[1]} = 0.30 + 0.80 = 1.10$, both passed by the ReLU. $z^{[2]} = 0.70 \cdot 0.10 - 0.60 \cdot 1.10 = -0.59$, $\hat{y} = \sigma(-0.59) = 0.36$, loss $-\log(1 - 0.36) = 0.44$. *Step 2.* $\delta^{[2]} = \hat{y} - y = 0.36$: the output was too high. *Step 3.* $\delta_1^{[1]} = 0.36 \times 0.70 \times 1 = 0.25$ and $\delta_2^{[1]} = 0.36 \times (-0.60) \times 1 = -0.21$: each sign follows the sign of the unit's outgoing weight. *Step 4.* Output weights: $0.36 \times 0.10 = 0.04$ and $0.36 \times 1.10 = 0.39$. Unit 1's weights: $0.25 \times (1, 1) = (0.25,\ 0.25)$. Unit 2's weights: $-0.21 \times (1, 1) = (-0.21,\ -0.21)$. Biases: $0.36$ (output), $0.25$ and $-0.21$ (hidden). One step with $\eta = 1$ lowers $\hat{y}$ to 0.11.

## The night before

Six things worth being able to state cold:

1. the three dissimilarity measures, what each ignores, and which are metrics;
2. what an eigenvector of the covariance is, in a sentence with no algebra;
3. ambient, embedding and intrinsic dimension, and what a $t$-SNE map cannot show;
4. what shattering measures and what CCGP measures, and why they pull against each other;
5. the three questions before trusting a score: above chance, held out, against its ceiling;
6. one update for three rules (learning rate $\times$ error $\times$ input), and the four steps of backpropagation.